Stablecoins at the Core of DeFi: Mechanisms, Risks, and the Search for Stability

A stablecoin is a cryptoasset engineered to track a target value (typically one US dollar). It underpins DeFi by providing a low-volatility unit of account, a collateral asset for lending and derivatives and a settlement asset for trading protocols[1]. Most decentralized exchanges quote pairs against dollar-pegged coins, lending protocols such as MakerDAO and Aave denominate loans in stablecoins, and on-chain derivatives often settle in them as well. This makes stablecoins as central to the crypto ecosystem as bank deposits are to traditional finance.

However, at system level, a stablecoin is not merely a token with a peg. It is an economic mechanism involving speculators, stablecoin users, governance token holders, oracle providers, and miners or validators. Its stability is an equilibrium outcome of their incentives and the protocol's rules[1]. The reading materials used in CS 194 develop a unified framework for analyzing these systems using tools from capital structure theory, currency peg models, and stochastic processes[1][2][3].

A useful high-level distinction is between custodial and non-custodial stablecoins[1]. **Custodial** designs hold off-chain reserves such as fiat deposits or short-term government debt, which are managed by a centralized issuer. Each token represents a claim on one unit of the reference asset. In the simplest case, reserves fully back liabilities, resembling a currency board or narrow bank; more complex designs allow fractional reserves, introducing maturity and credit transformation similar to commercial banks[1]. Price stability is enforced through redemption arbitrage: when the token trades below $1, arbitrageurs buy and redeem; when it trades above, they mint and sell. Risk arises primarily from counterparty, asset-quality, and regulatory exposure.

Non-custodial stablecoins instead implement the capital structure on-chain. Klages-Mundt et al. decompose these systems into functional components: primary value (collateral), risk-absorbing positions, stablecoin holders, issuance and deleveraging rules, governance, oracles, and miners/validators[1]. Primary value may be either *exogenous collateral* such as ETH, whose price is determined by external markets, or *endogenous collateral* such as governance tokens, whose value is reflexive to the system's success, or *implicit collateral* in seigniorage-style designs[1]. Risk is borne either directly by collateralized borrowers (vault owners) or indirectly by equity-like governance token holders, who can be diluted to recapitalize shortfalls.

Issuance and deleveraging rules are central. In an over-collateralized lending model inspired by Maker's Dai[2], agents lock collateral and borrow stablecoins subject to a collateralization ratio β. If the collateral's value falls so that the ratio is violated, a liquidation is triggered. Liquidations sell collateral for stablecoins, which are then burned to reduce supply. This gives speculators a leveraged long position in the collateral asset and a short position in the stablecoin.

Governance, oracles, and MEV form additional layers of risk[1][3]. Governance can set parameters (fees, collateral ratios, auction rules) and seize collateral in extreme cases. Oracles supply price data that determine whether positions are under-collateralized. Miners or validators can reorder transactions to capture arbitrage or liquidation profits (MEV). These components are integral to the mechanism, if mis-aligned, they can destabilize the system even if the core peg rule is sound.

In order to more quantitatively examine the various performances of stablecoins, "While Stability Lasts" formalizes the dynamics of an over-collateralized stablecoin using a *stochastic model*[2]. The system has a risky collateral asset with exogenous price process X_t and an endogenous stablecoin price Z_t. Stablecoin supply L_t is chosen by risk-neutral speculators who compare the benefits of leverage against expected liquidation costs. Demand for the stablecoin is modelled as imperfectly elastic in dollars, so that market clearing implies $Z_t = D/L_t$ for some dollar demand D[2].

Two constraints govern behavior: the collateral constraint (liquidation when value falls below βL_t and the market-clearing relationship between price and supply. Under mild assumptions on asset returns and fees, the authors show that speculators' optimization is concave and yields well-defined best responses[2]. Within this framework they identify a *stable domain* in which deviations from the peg are rare and variance is bounded, and an *unstable domain* in which the stochastic process can be driven far from the peg by deleveraging alone.

A key phenomenon is the **deleveraging spiral**[2][3]. When the collateral price crashes, many positions breach their collateral requirement and are liquidated, which are paid with stablecoins that are burned, and which decreases supply. If dollar demand D is slow to adjust (because stablecoins are locked in DeFi contracts or users face frictions), then reducing supply forces the stablecoin price *above* $1, since $Z_t = D/L_t$ increases as L_t falls. Because of this premium, liquidations become more costly in collateral terms, and voluntary deleveraging requires repurchasing stablecoins at an inflated price. The system can enter a deflationary spiral in which collateral is drained rapidly while the stablecoin trades at a premium.

Black Thursday in March 2020 provides an empirical case study. A sharp drop in ETH price, combined with Ethereum congestion, caused Maker liquidations to clear at near-zero bids, leaving a system-wide collateral shortfall. Dai traded at substantial premiums as users scrambled to acquire the stablecoin to close positions[2][3]. Earlier models of "stablecoin attacks" drew similar lessons from events in bitUSD, Steem Dollars, and NuBits: levered stablecoins can fail through endogenous feedback even without explicit protocol bugs[3].

Stablecoins 2.0 complements this stochastic analysis with a capital-structure view of *incentive security*[1]. If governance token holders can expropriate collateral, by voting to change rules once they control a quorum for instance, there may exist parameter regions where such an attack is strictly profitable. Anticipating this, rational vault owners and stablecoin users may never enter the system. The design problem is thus to choose governance rules and legal constraints so that the expected present value of honest operation outweighs any one-shot theft, aligning participation incentives for all agents[1].

The premise of the discussion of the risks of these stablecoins is that the importance of stablecoins to DeFi is self-evident, not only because they represent "stable money" but because they act as *universal collateral* and *numéraire* for an entire ecosystem. Most on-chain lending is denominated in stablecoins, and they are the base asset for many derivatives and synthetic positions. They also increasingly serve as cross-border payment rails and as a dollar substitute in high-inflation economies[4][5]. This centrality makes their failure become a source of systemic risk, which can propagate through liquidation cascades, oracle dependencies, and MEV-driven arbitrage across multiple protocols simultaneously[3].

The reading materials highlight several open design challenges. For example, over-collateralized designs often operate with thin buffers: once the collateral process leaves a "good" regime, there may simply not be enough risk-absorbing capital to keep the peg[1][2]. Explicit buffers, such as fee-funded reserves, mezzanine tranches or insured backstops, can enlarge the stable domain but may push designs closer to custodial models. What's more, robustness of oracles and governance must be analyzed jointly with the peg mechanism, rather than treated as exogenous infrastructure, because mis-aligned incentives in these layers can undermine any stability guarantees[1][3]. Additionally, liquidation design needs to be MEV-aware. When miners or searchers can profitably manipulate auction prices or timing, they may amplify crises instead of providing stabilizing liquidity[3].

In conclusion, according to the readings, stablecoins have become indispensable infrastructure for DeFi, but their stability is far from guaranteed. The theoretical frameworks surveyed here show that price

stability in permissionless systems depends on a delicate alignment of incentives across speculators, users, governors, oracle providers, and miners, as well as on sufficient buffers against downside shocks[1][2][3]. Over-collateralization alone does not prevent deleveraging spirals, and under realistic demand frictions, liquidations can drive the stablecoin *above* its peg precisely when the system is most fragile[2]. At the same time, governance structures that are too powerful introduce profitable attack vectors that can deter participation altogether[1][3].

Understanding these mechanisms is essential if stablecoins are to continue serving as the backbone of an open financial infrastructure. The models developed in the CS 194 readings provide a foundation for future designs that better balance capital efficiency, decentralization, and robustness, bridging the gap between traditional financial theory and the realities of on-chain markets.

References

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